Causes and risk factors: Obesity is due to genetic factors, physical inactivity, eating habits (overeating) and endocrine factors. Obesity is a positive risk factor in development of hypertension, diabetes, gall bladder disease, coronary heart disease and arthritis.

21.9.1 Prevention and Control of Obesity

Diet Management: Low calorie, normal protein, vitamins and mineral, restricted carbohydrate and fat, high fiber diet can prevent overweight. Calorie restriction for weight reduction is safe and most effective.

Physical exercise: A low calorie diet accompanied by moderate exercise will be effective in causing weight loss. Meditation, yoga and physical activity can also reduce stress related to overeating.

21.10 Heart Disease

Cardiovascular disease (CVD) is associated with diseases of the heart and blood vessels. **Coronary heart disease** (CHD) is the most common form and is caused by **deposition of cholesterol in the blood vessels**.

It usually develops slowly over many years beginning from childhood, they may form a fatty streak to a fibrous complicated **plaque**. It leads to the narrowing of blood vessels leading to **atherosclerosis** in the large and medium sized arteries that supply the heart muscle with oxygen. It leads to sudden **ischemia** (deficient blood supply to heart muscle) and **myocardial infarction** (death of the heart muscle tissue).

More to Know

Desirable level for blood cholesterol should be less than 200 mg/dl for Indians. The risk of coronary heart disease increases slowly as blood cholesterol levels increases from 200 to 300 mg/dl.

Risk factors: Hypercholesterolemia (High blood cholesterol) and high blood pressure (Hypertension) are the major causes and contributing factors for heart disease and if untreated may cause severe damage to brain, kidney and eventually lead to stroke.

Causes: Heredity (family history), diet rich in saturated fat and cholesterol, obesity, increasing age, cigarette smoking, emotional stress, sedentary lifestyle, excessive alcohol consumption and physical inactivity are some of the causes.

Symptoms: Shortness of breath, headache, tiredness, dizziness, chest pain, swelling of leg, and gastrointestinal disturbances.



HDL (High Density Lipoprotein) or "**good**" cholesterol lowers risk of heart disease while **LDL** (Low Density Lipoprotein) or "**bad**" cholesterol increases risk of heart disease.

21.10.1 Prevention and Control of Heart Disease

Diet management: Reduction in the intake of calories, low saturated fat and cholesterol rich food, low carbohydrates and common salt are some of the dietary modifications. Diet rich in polyunsaturated fatty acids (PUFA) is essential. Increase in the intake of fibre diet, fruits and vegetables, protein, minerals and vitamin are required.

Physical activity: Regular exercise, walking and yoga are essential for body weight maintenance

Addictive substance avoidance: Alcohol consumption and smoking are to be avoided.

Activity 3

Prepare a chart showing the food items which are preferable and which should be avoided to prevent high blood pressure and heart disease. Apart from diet what are the other lifestyle modifications to be followed to manage this condition.

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